

DISTRIBUTED COMPUTING

Time: Three hours

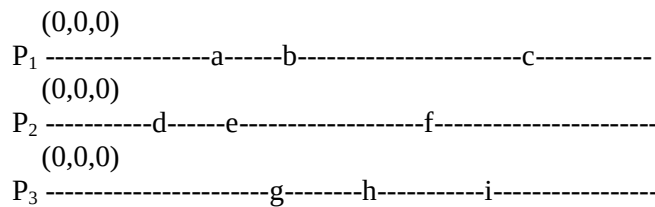
Full Marks: 100

Attempt **ALL FIVE** questions

1. (CO1) Assume a distributed system with three sites, all of which are equipped with vector clocks that are initialized to (0, 0, 0). Assume further that each letter in the diagram represents an event as explained in the accompanying table.

Suppose all messages are broadcast and the Birman-Schiper-Stephenson protocol is used to maintain causal ordering of messages. In this protocol, a message is delivered to a process only if the message preceding it immediately has been delivered to the process. Otherwise, the message is buffered.

Show all vector clock updates in the three sites as a result of these message sending activities. Please mark all your clock updates on the diagram.

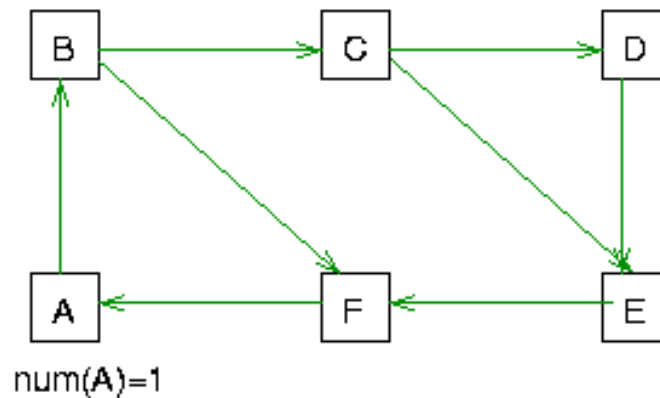


LABEL | EVENT

a	M ₁ arrives at P ₁
b	P ₁ broadcasts M ₃ to P ₂ & P ₃
c	M ₂ arrives at P ₁
d	P ₂ broadcasts M ₁ to P ₁ & P ₃
e	P ₂ broadcasts M ₂ to P ₁ & P ₃
f	M ₃ arrives at P ₂
g	M ₂ arrives at P ₃
h	M ₃ arrives at P ₃
i	M ₁ arrives at P ₃

2. (CO4) Answer the following questions.

- What is the AND Model of deadlock in a distributed system? How can it be detected from a Wait For Graph (WFG)?
- What is the OR Model of deadlock in a distributed system? How can it be detected from a Wait For Graph (WFG)?
- Describe the Chandy-Misra-Haas algorithm for deadlock detection in a distributed system. Assume AND model.
- Explain the working of Chandy-Misra-Haas algorithm on the following WFG. Assume OR model.



$$(2+1)+(2+1)+8+6$$

3. (CO3) Answer the following questions.

- Consider Ricart-Agarawala's mutual exclusion algorithm being run on a synchronous system with maximum message delay T for any link. Processing time at each node is negligible. The clocks are not synchronized but you can assume that they have zero drift. Is it possible to improve the message complexity (no. of messages per CS execution) of the algorithm by modifying it in some way? Justify your answer clearly
- Given a distributed system with three processes P_1 , P_2 , and P_3 , in three different sites where mutual exclusion is enforced using Ricart-Agrawala's algorithm. Consider the scenario below and list in detail all the events that follow until both processes complete their critical sections.
 - Both P_1 and P_2 are requesting for critical section while P_3 has no need at this time.
 - The timestamp of P_1 's request is 47, and that of P_2 is 32.

$$8+(4+8)$$

4. (CO5) Answer the following questions.

- Describe NFS architecture and NFS V-Node.
- How is *look up* implemented in NFS V3?
- Discuss *open delegation* method in NFS v4.
- Discuss *compound procedures* method in NFS v4.
- What is meant by UNIX file semantics and session semantics?

$$(2+2)+4+4+4+(2+2)$$

5. (CO6)
- (a) Consider the 2-phase commit protocol. Clearly explain how the coordinator recovers if it has failed (i) before writing the commit record in the log but after receiving AGREED messages from all other processes, and (ii) after writing the commit record in the log and getting back some (but not all) acknowledgements from the other processes.
 - (b) Prove that two phase locking is a sufficient condition for serializability. Show that it is not necessary (you may use a counter example).

(3+3)+(7+7)